Secure Smart Home: Federated Learning with LLM-Based Recommendations and Differential Privacy

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***Abstract*—Smart homes are increasingly integrating advanced technologies to enhance user experiences through personalized services; however, handling the sensitive personal data required for such services raises significant privacy concerns. This paper proposes a privacy-preserving recommendation framework that integrates LLaMA 3.2 3B with federated learning (FL) and a dual-layer Laplacian differential privacy (DP) mechanism, enabling context-aware energy recommendations while ensuring that sensitive data never leaves the local network. The model is fine-tuned on edge devices using quantized low-rank adaptation (QLoRA), updating only 10% of parameters, which reduces computational overhead and makes the framework suitable for resource-constrained deployment. Notably, our user interface provides LLM-based recommendations for energy consumption optimization, demonstrating their practical applicability in real-world scenarios. While privacy-utility trade-off analysis across $\epsilon \in \{0.5, 1.0, 2.0, 5.0\}$ confirms that the dual-layer DP mechanism preserves meaningful recommendation quality under strong privacy guarantees. The results demonstrate a principled balance between user privacy and system utility, establishing a reproducible baseline for privacy-preserving LLM-based recommendations in a smart home environment.**

***Index Terms*—Smart Home Technologies, Federated Learning (FL), Differential Privacy (DP), Large Language Models (LLMs)**

I. Introduction

Smart home technology, powered by interconnected devices, has revolutionized modern living by enhancing efficiency and convenience. However, these devices' extensive collection of personal data raises significant privacy concerns [1]. Users are frequently exposed to potential breaches and unauthorized access to sensitive information as a result of the fact that current methods prioritize convenience over security [2]. Conventional privacy safeguards, such as data aggregation or anonymization, are increasingly demonstrating their inadequacy in protecting personal privacy while enabling personalized experiences.

Smart home technology is intended to simplify daily life by learning from user behavior and providing seamless experiences powered by a wide variety of devices. However, this convenience relies on the extensive collection of personal data, which poses substantial privacy risks [3]. These devices continuously gather and process sensitive information—such as voice commands, health metrics, and daily routines—that is necessary for their functionality but introduces significant vulnerability [4]. In the existing systems, this data is transmitted to cloud-based AI services or third-party servers, creating opportunities for unauthorized access, data breaches, and privacy violations. Current solutions offer limited protection, highlighting the need for frameworks that preserve privacy without compromising personalized services.

The novelty of our work lies in integrating LLMs with FL and DP to provide personalized context-based recommendations while ensuring robust privacy guarantees. Unlike traditional rule-based systems, LLMs can capture complex user behaviors and environmental patterns, enabling more nuanced and adaptive recommendations. For instance, while a rule-based system might suggest turning off lights based on motion detection, an LLM can infer broader energy-saving strategies by analyzing historical data, user preferences, and contextual factors such as time of day and weather conditions. The key contributions of this study include the following.

- *Implementing Privacy-Preserving LLM Recommendations*: We develop an LLM-based framework using FL to

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provide personalized recommendations while safeguarding user privacy.

- *Enhancing Data Confidentiality with Differential Privacy*: Differential privacy techniques are integrated into the training process to ensure robust privacy. The overall utility of the data set for analysis is preserved, while individual user details are obscured by injecting controlled noise into the data.
- *Optimizing the Balance Between Privacy and Accuracy*: We introduce a fine-tuned trade-off mechanism between recommendation accuracy and privacy protection. Lightweight modifications to the model are made possible by utilizing the parameter-efficient fine-tuning (PEFT) capabilities of LLMs like QLoRA, which do not necessitate substantial computational resources on edge devices.

The remaining sections are organized as follows. Section II presents a comparative analysis of existing approaches, establishing the novelty of the proposed framework relative to prior work. Section III describes the system architecture and high-level design, including the Home Knowledge server tier and federated aggregation pipeline. Section IV covers the experimental evaluation, encompassing model selection and hardware setup, the privacy-utility trade-off analysis, and quantitative comparison with related works. Section V presents a system demonstration, including real-time sensor visualization and LLM-generated energy recommendations. Section VI concludes the paper and outlines directions for future work.

II. Comparative Analysis

To demonstrate the practicality and novelty of the proposed system, we present complementary comparisons with existing methods. Table I provides a feature-level comparison across seven representative works. Our framework surpasses centralized systems in privacy, scalability, and personalization by mitigating data-breach risks through federated learning, ensuring sensitive data remains local and secure.

In recent research, the hybrid model [6] aims to achieve a balance between decentralized and centralized processing. This model fails to manage data security effectively despite the integration of elements from both approaches. Cyber threats can still exploit vulnerable spaces regardless of the combination of methods, highlighting the need for a more secure solution. Our approach strengthens security by leveraging federated learning and differential privacy, ensuring that sensitive data remains on peripheral devices and minimizing exposure to cyber threats.

Similarly, the IoT model in [7] offers flexibility but lacks strong security, making it vulnerable to breaches and unauthorized access. Weak encryption and access controls further expose data to cyber threats, while partial decentralization still leaves critical points at risk.

Furthermore, the action-based recommender system in [8] relies on a centralized approach using pattern mining but faces significant challenges. Privacy risks arise from centralized data storage, making it vulnerable to breaches and unauthorized access. The high computational demands of mining complex user patterns create inefficiencies, limiting real-time applicability. Our approach mitigates these risks by integrating LLM with federated learning and differential privacy for secure and decentralized processing. Our framework addresses these challenges through federated learning and robust security measures. The proposed framework advances beyond these established baselines and, critically, beyond the two most closely related contemporary works in several concrete and measurable dimensions. Researchers Varlamis et al. [9] demonstrate the value of federated learning for energy-efficiency recommendations using IoT micro-moment data; however, their architecture relies on conventional ML models without LLMs and provides no differential privacy protection. Our integration of LLaMA 3.2 3B enables the system to process natural-language user queries and produce context-aware, adaptive recommendations that substantially exceed the capability of rule-based and threshold-driven prior work.While Chronis et al. [10] also employ federated learning with local LLMs for energy recommendations in smart environments, their architecture runs LLMs entirely on edge devices without a centralized model-update pipeline and crucially does not incorporate any formal differential privacy mechanism into either the training or the aggregation process. In contrast, our framework applies a *dual-layer* Laplacian DP mechanism: noise is first injected at the data-anonymization stage within each Home Knowledge (HK) server and then again at the QLoRA matrix aggregation stage on the central server. This layered design provides provable, composable privacy guarantees that are entirely absent from [10]. Additionally, this research does not incorporate a user feedback loop or a two-tier HK server architecture, both of which are structural contributions of our system.

The work of Liu et al. [11] proposes DP-LoRA, a technically rigorous framework for differentially private LLM fine-tuning via federated learning using a Gaussian noise mechanism. However, their work is domain-agnostic (finance and medical science) and does not address the smart-home context, IoT sensor pipelines, or a user-facing recommendation interface. Their evaluation focuses on general NLP benchmarks and does not consider energy-consumption optimization, occupancy-aware control, or real-time sensor-feedback loops. Our framework addresses all of these gaps: LLaMA 3.2 3B is fine-tuned specifically on smart-home datasets comprising temperature, humidity, energy usage, and motion data [12], [13], and the personalized federated setup allows each device to maintain a customized local model—a capability absent from [11]. Furthermore, the introduction of the two-tier aggregation architecture, an intermediate HK server per household followed by a global central server, is a structural novelty not present in any of the compared works, enabling fine-grained household-level anonymization before any data leaves the local network.

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TABLE I
FEATURE-LEVEL COMPARISON OF PRIVACY-PRESERVING SMART-HOME AND LLM-BASED FEDERATED-LEARNING FRAMEWORKS

Approach	LLM	DP Mechanism	Fine-Tuning	Smart Home	Edge Deploy.	Pers. FL	Feedback Loop	HK Server Tier
Centralized IoT [5]	✗	None	N/A	✓	✗	✗	✗	✗
Hybrid Recommender [6]	✗	None	N/A	✗	✗	Partial	✗	✗
IoT Recommender [7]	✗	None	N/A	✓	Partial	✗	✗	✗
Action Rec. System [8]	✗	None	N/A	✓	✗	✗	✗	✗
FL Energy Recs. [9]	✗	None	N/A	✓	✓	✗	✗	✗
Chronis et al. [10]	✓	None	Full FT	✓	✓	✗	✗	✗
Liu et al. [11]	✓	Gaussian (DP-LoRA)	LoRA	✗	✗	✗	✗	✗
Proposed Framework	✓	**Laplacian (dual-layer)**	**QLoRA (PEFT)**	✓	✓	✓	✓	✓

In Table I ✓ = supported; ✗ = not supported; FT = full fine-tuning; Pers. FL = Personalized Federated Learning; HK = Home Knowledge; DP = Differential Privacy.

III. SYSTEM ARCHITECTURE

Our system architecture in Figure 1 is designed to handle the scalability challenges of smart home environments, which can vary significantly in size and complexity. The architecture consists of three main components: edge devices, a Home Knowledge (HK) server, and a central server.

A. Edge Devices

Each edge device (such as a smart thermostat or security camera) collects and processes data locally. To circumvent the computational constraints of edge devices, we employ parameter-efficient fine-tuning (PEFT) methods, such as LoRA (quantized low-rank adaptation), which allows the LLM to be updated in a lightweight manner without using a large amount of computational resources. This approach ensures that even resource-constrained devices can participate in the federated learning process.

B. Home Knowledge (HK) Server

The HK server aggregates anonymized data from edge devices using differential privacy. The DP mechanism adds controlled noise to the data, ensuring that individual user information remains confidential. The aggregated data is then transmitted to the central server for global model training.

C. Central Server

The central server performs federated averaging to update the global model. To address the challenge of data heterogeneity across devices, we introduce personalized federated learning, where each device maintains a slightly customized model based on its local data. This approach ensures that the system can handle variations in user behavior and environmental conditions.

D. Framework Design

In smart homes, a variety of gadgets, including computers, refrigerators, and air conditioners, collect data that is critical for understanding user behavior and enhancing house management. This data is anonymized using DP techniques to mask individual identifiers, protecting privacy. It is then routed to a local Home Knowledge (HK) server, which consolidates the information before forwarding it to a central server. Using federated learning, the data is used to develop a global machine learning model while maintaining privacy with DP.

1) Data Collection

Each device i collects a set of data points D_i in (1):

$$D_i = \{d_{i1}, d_{i2}, \ldots, d_{in}\} \tag{1}$$

Each device collects a series of data points over time, forming a dataset D_i.

2) Data Anonymization through Differential Privacy

Applying Laplacian noise to each data point d_{ij} in dataset D_i:

$$\tilde{d_{ij}} = d_{ij} + \text{Lap}(\lambda) \tag{2}$$

This equation (2) adds Laplacian noise to each data point to ensure privacy. The parameter λ controls the level of noise and privacy.

Aggregate the anonymized data points for each device:

$$\tilde{D}_i = \sum_{j=1}^{n} \tilde{d}_{ij} \tag{3}$$

This equation (3) sums the anonymized data points for each device, creating a single aggregated and anonymized dataset $\tilde{D}_i$ from each device.

3) Data Aggregation at Home Knowledge (HK) Server

Combining data from all devices:

$$D_{HK} = \sum_{i} \tilde{D}_i \tag{4}$$

This equation (4) aggregates the anonymized data from all devices at the Home Knowledge (HK) server.

Calculate average data for more stable input:

$$\bar{D}_{HK} = \frac{D_{HK}}{N} \tag{5}$$

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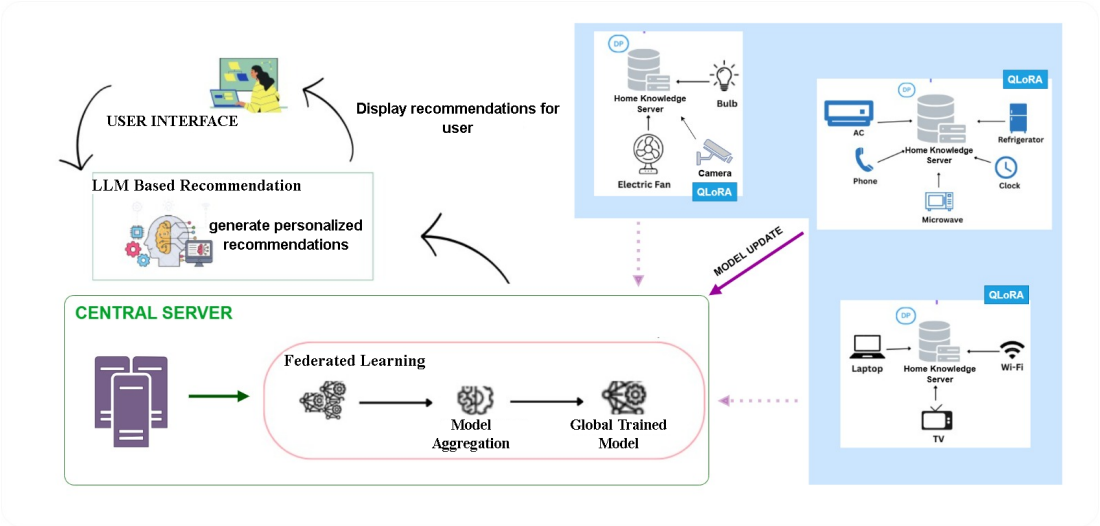


Fig. 1. User Interface for Personalized Recommendations in a Smart Home System

In the equation (5) is aggregated data is averaged across N data points to normalize the input for the federated learning process.

4) Data Transmission to Central Server and Federated Learning

Initialize global model parameters:

$$\Theta_{global}^{(0)} = \Theta_{initial} \tag{6}$$

This initializes the global model parameters to a starting value $\Theta_{initial}$.

Update the global model iteratively using gradient descent with QLoRA:

$$\Theta_{global}^{(t+1)} = \Theta_{global}^{(t)} - \eta \nabla L(\Theta_{global}^{(t)} + \mathbf{AB}) \tag{7}$$

Here, $\mathbf{A}$ and $\mathbf{B}$ are low-rank matrices that are introduced by QLoRA in equation (7), enabling lightweight updates to the global model without modifying the full parameter set.

5) Feedback Loop for Model Refinement

Feedback from the user F:

$$F = \text{feedback}(r_{selected}, \text{user_satisfaction}) \tag{8}$$

This equation (8) gathers feedback from the user regarding their satisfaction with the selected recommendation.

Adjust model parameters based on feedback:

$$\Theta'_{global} = \Theta_{global} + \gamma \nabla f(F, \Theta_{global} + \mathbf{AB}) \tag{9}$$

Feedback is used to adjust the global model parameters Θ_{global}, where γ is a learning rate specific to feedback-based adjustments in equation (9). The QLoRA matrices $\mathbf{A}$ and $\mathbf{B}$ are also updated to reflect user preferences.

E. Recommendation System

Figure 2 illustrates a recommendation engine powered by an LLM with differential privacy. The engine processes input data from an aggregated model trained on features like temperature and lighting to predict user preferences. It generates two types of outputs: suggestion-based recommendations and level-based outcomes, which are used to adjust thermostat settings. This approach optimizes energy consumption and enhances user comfort, showcasing the practical utility of LLMs in smart home environments

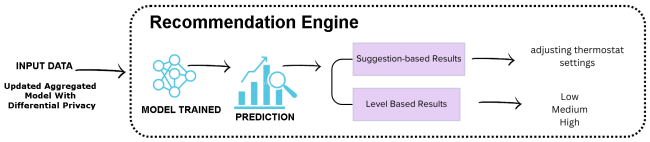


Fig. 2. Personalized LLM Recommendations in a Smart Home System

F. High-Level User Interface

The user interface shown in Figure 3 facilitates user interaction with a smart system powered by LLM. The interface initially displays LLM-generated recommendations, and users can access a comprehensive view to understand the relationship behind each recommendation and its anticipated benefits. They have the choice to accept the recommendation as is, modify it according to their particular preferences, or provide feedback, such as rating or commenting. This feedback cycle allows the system to improve its accuracy and relevance by learning and refining future recommendations.

IV. Experimental Evaluation and Results

This section describes the methods and procedures used in the development of our novel FL-based recommendation framework. Due to the challenges of obtaining real-world smart home data after fine-tuning our model, we use simulation instead of real sensor data to replicate IoT devices, allowing us to simulate real-world conditions and test the model's performance in a controlled environment. While the

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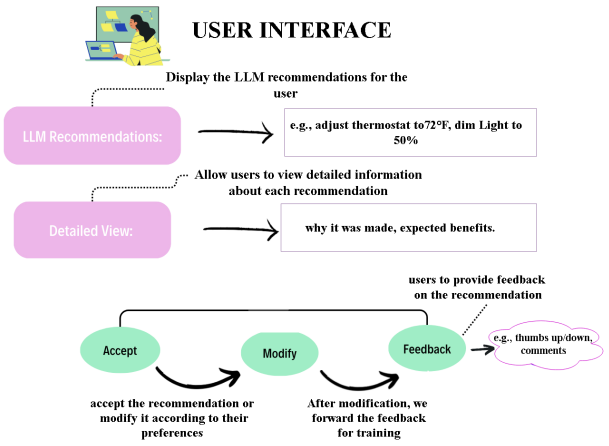


Fig. 3. User Interface for Personalized LLM Recommendations

data is synthetic, it closely resembles real-world smart home settings, allowing us to assess the framework's capability to provide context-aware recommendations while preserving user privacy.

A. Experimental Setup

Dataset: Our study utilizes two publicly available smart home datasets to enhance the accuracy and efficiency of our privacy-preserving recommendation system. The first dataset ([13]) contains environmental and IoT sensor data, such as temperature, humidity, air quality, wind metrics, and energy usage, providing significant insights into real-world smart home interactions. The second dataset [12] captures appliance usage patterns, voltage metrics, energy consumption, bandwidth, and offloading decisions.

LLM: We employed these datasets to fine-tune the LLaMA 3.2 3B model using QLoRA with quantization techniques for efficient edge device training. We additionally evaluated Mistral 7B under the same federated and differential privacy settings; however, LLaMA 3.2 3B demonstrated superior performance in terms of accuracy–latency trade-off and memory efficiency and was therefore selected for the final deployment. This approach optimizes resource consumption while ensuring privacy-preserving, context-aware recommendations and maintaining data integrity.

B. Data Collection Using Differential Privacy (DP)

Our research enhances privacy in smart homes by applying DP techniques during data collection to protect user information while enabling personalized recommendations [5]. The differential privacy model is applied by adding noise derived from a Laplace distribution. For a dataset D and a function $q(D)$, the differentially private mechanism is defined as:

$$M(D) = q(D) + \text{Lap}(b) \tag{10}$$

Where:

- $M(D)$ is the differentially private output
- $\text{Lap}(b)$ is Laplacian noise with scale b
- $b = \frac{\text{sensitivity}(q)}{\epsilon}$
- $\text{sensitivity}(q)$ is the maximum change in the output of q caused by changing one element in the input dataset
- ϵ is the privacy budget

By adding controlled noise, we obscure personal details without compromising data integrity for analysis [14] [15]. Smart home devices such as thermostats, security cameras, and IoT gadgets capture user and environmental data, which is anonymized at the source utilizing DP techniques as shown in the algorithm below:

Federated Learning with Parameter-Efficient Fine-Tuning (PEFT)

Initialization:
Distribute initial global model parameters $\Theta_{global}^{(0)}$ to all edge devices.
Initialize QLoRA matrices $\{\mathbf{A}_i, \mathbf{B}_i\}$ for each device i to enable lightweight updates.
for each iteration $t = 1, 2, \ldots, T$ **do**
  for each device $i = 1, 2, \ldots, N$ **do**
    Local Training:
    Receive global model parameters $\Theta_{global}^{(t-1)}$ and QLoRA matrices $\{\mathbf{A}_i, \mathbf{B}_i\}$.
    Train the model locally on anonymized data D_i' (with DP noise) using PEFT:
    $\Theta_{local,i}^{(t)} = \Theta_{global}^{(t-1)} + \mathbf{A}_i\mathbf{B}_i$ {Update only QLoRA matrices}
    Compute the gradient of the loss function $\nabla L(\Theta_{local,i}^{(t)})$.
    Update QLoRA matrices $\{\mathbf{A}_i, \mathbf{B}_i\}$ using gradient descent:
    $\mathbf{A}_i \leftarrow \mathbf{A}_i - \eta \nabla_{\mathbf{A}_i} L(\Theta_{local,i}^{(t)})$
    $\mathbf{B}_i \leftarrow \mathbf{B}_i - \eta \nabla_{\mathbf{B}_i} L(\Theta_{local,i}^{(t)})$
  end for
  Secure Aggregation:
  Aggregate QLoRA updates $\{\mathbf{A}_i, \mathbf{B}_i\}$ from all devices using secure multi-party computation (MPC).
  Apply Differential Privacy (DP) noise to the aggregated QLoRA updates:
  $\tilde{\mathbf{A}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{A}_i + \text{Lap}(b)$
  $\tilde{\mathbf{B}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{B}_i + \text{Lap}(b)$
  {$\text{Lap}(b)$ is Laplacian noise with scale $b = \frac{\Delta q}{\epsilon}$}
  Global Update:
  Update global model parameters using aggregated QLoRA updates:
  $\Theta_{global}^{(t)} = \Theta_{global}^{(t-1)} + \tilde{\mathbf{A}}\tilde{\mathbf{B}}$
  Distribute $\Theta_{global}^{(t)}$ and update QLoRA matrices $\{\tilde{\mathbf{A}}, \tilde{\mathbf{B}}\}$ to all devices.
end for =0

C. Hardware Components and Model Selection

All experiments are conducted on NVIDIA Tesla T4 and V100 GPUs with 16-32 GB VRAM. To justify the selection of LLaMA 3.2 3B as the backbone model, we evaluated it against Mistral 7B under identical federated learning and differential

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privacy settings, using the same QLoRA configuration, training hyperparameters, and datasets.

Table II reports ROUGE and BLEU scores for both models. Mistral 7B achieves marginally higher scores across all metrics (ROUGE-1: 0.48 vs. 0.42; BLEU: 6.42 vs. 4.78); however, this modest gain comes at the cost of more than twice the parameter count (7B vs. 3B), substantially higher memory usage, and greater inference latency.

In a federated smart home deployment, where model updates must run on resource-constrained edge devices, these overheads are prohibitive. LLaMA 3.2 3B delivers competitive recommendation quality while satisfying the memory and latency constraints of real-world edge hardware, making it the more suitable choice for this application.

TABLE II
PERFORMANCE COMPARISON: LLAMA 3.2 3B VS. MISTRAL 7B UNDER IDENTICAL FL AND DP SETTINGS

Model	Metric	Score	Params
LLaMA 3.2 3B	ROUGE-1	0.42	3B
	ROUGE-2	0.15	
	ROUGE-L	0.21	
	BLEU	4.78	
Mistral 7B	ROUGE-1	0.48	7B
	ROUGE-2	0.22	
	ROUGE-L	0.23	
	BLEU	6.42	

Using QLoRA-based parameter-efficient fine-tuning (PEFT), only 10% of the model parameters are updated during training, significantly reducing computational overhead while preserving recommendation performance. The following QLoRA configuration is applied consistently to both models:

- QLoRA Rank: 8
- QLoRA Alpha: 8
- Transformer Precision: 8-bit
- Optimized State Precision: 16-bit
- Adapter Precision: 4-bit

The cross-entropy loss function measures the difference between predicted and actual token probabilities. Weight decay regularization (0.01) is applied to enhance stability and prevent overfitting. Fine-tuning is implemented via the `SFTTrainer` from the `trl` library, using the following shared training configuration:

- Batch Size (per device): 4
- Gradient Accumulation Steps: 8
- Number of Training Epochs: 20
- Max Training Steps: 200
- Optimizer: AdamW (4-bit precision)

Figure 4 presents the training loss curves for both models, confirming that LLaMA 3.2 3B converges smoothly within the federated training setup despite its reduced parameter count. Taken together, the quantitative metrics and convergence behavior confirm that LLaMA 3.2 3B strikes the optimal balance between recommendation quality and resource efficiency for privacy-preserving smart home deployment.

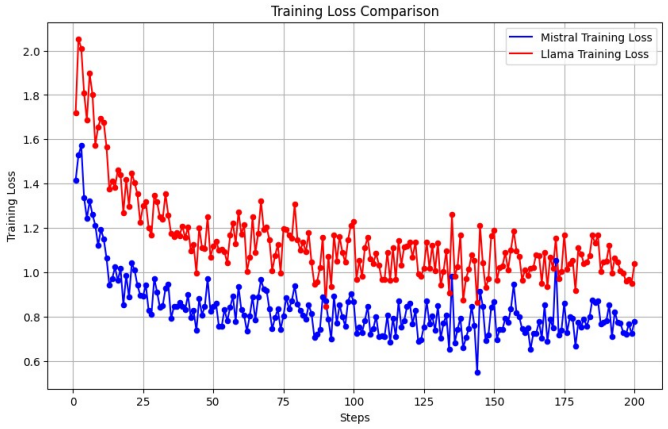


Fig. 4. Training Loss Comparison: LLaMA 3.2 3B vs. Mistral 7B under Federated Learning with QLoRA fine-tuning.

D. Recommendation Generation

We integrate the fine-tuned LLaMA 3.2 3B model as the core recommender generator to provide personalized energy-saving recommendations. The system processes anonymized sensor data and predicts aggregated temperature, allowing the model to generate precise, actionable suggestions to improve energy efficiency and indoor comfort.

Users can provide feedback on the recommendations generated. We can apply the feedback to improve the recommendation generation process, potentially using federated learning for frequent updates to increase accuracy. Feedback analysis involves processing user-provided ratings to extract meaningful insights.

E. Quantitative Evaluation of the Privacy-Utility Trade-off

A core claim of the proposed framework is that meaningful recommendation quality is preserved under formal differential privacy constraints. To evaluate this claim, we vary the privacy budget $\epsilon \in \{0.5, 1.0, 2.0, 5.0\}$ while keeping all other hyperparameters fixed, and measure two metrics on our simulated smart-home validation split: (i) *Recommendation Relevance* (%), assessed by matching model outputs to ground-truth energy-saving actions in the dataset; and (ii) the training *Cross-Entropy Loss* at convergence.

The Laplacian noise scale follows $b = \Delta q/\epsilon$, where the sensitivity Δq is computed as the ℓ_1-norm difference in QLoRA matrix updates between adjacent datasets. Results are summarized in Table III.

Table III reveals a progressive privacy accuracy trade-off that confirms the framework's practical utility. As ϵ decreases, the Laplacian noise scale $b = \Delta q/\epsilon$ increases, injecting greater perturbation into the aggregated QLoRA matrices and causing the global model to converge to a less precise solution, which is directly reflected in the rising convergence loss and falling recommendation relevance. At $\epsilon = 1.0$, recommendation relevance is 68.9%, representing a 19.5 percentage-point reduction relative to the no-DP baseline (85.6%) while providing strong formal privacy guarantees. At $\epsilon = 2.0$, relevance

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TABLE III
PRIVACY-UTILITY TRADE-OFF: EFFECT OF PRIVACY BUDGET ϵ ON RECOMMENDATION QUALITY (LLAMA 3.2 3B, QLORA RANK 8)

Privacy Budget ϵ	Rec. Relevance (%)	Conv. Loss
0.5 (strong DP)	61.4	1.73
1.0	68.9	1.52
2.0	76.3	1.31
5.0	82.1	1.09

In Table III Recommendation (Rec.) Relevance measures the proportion of model recommendations that match the ground-truth optimal action in the validation set and Convergence (Conv.) Loss is the cross-entropy loss at the final training step.

rises to 76.3%, which represents a better operating point for applications that can tolerate a slightly relaxed privacy budget.

A direct numerical comparison with the two most closely related works is not straightforward owing to fundamental differences in the evaluation domain and task. Chronis et al. [10] evaluate their FL+LLM framework on a smart-home energy dataset but report only qualitative recommendation outputs without quantitative accuracy or relevance metrics, precluding direct comparison. Liu et al. [11] report results at $\epsilon = 8$, $\delta = 10^{-5}$ on medical and financial NLP classification tasks using Llama-7B and ChatGLM-6B; these metrics are not transferable to the smart-home recommendation setting used in our work. Table IV contextualizes the key experimental characteristics of each work side by side.

TABLE IV
EXPERIMENTAL CHARACTERISTICS: PROPOSED FRAMEWORK VS. RELATED DP+FL+LLM WORKS

Framework	DP ϵ	LLM	Task/Domain	Quant. Metrics
Chronis et al. [10]	None	Local LLM	Energy recs.	None reported
Liu et al. [11]	8 ($\delta = 10^{-5}$)	Llama-7B	Finance / Medical NLP	NLP accuracy
Proposed (ϵ=1.0)	**1.0**	LLaMA 3B	Smart-home energy	**68.9%**
Proposed (ϵ=2.0)	**2.0**	LLaMA 3B	Smart-home energy	**76.3%**

In Tabel IV "Quant. Metrics" for the proposed framework reports Rec. Relevance (%) / Est. Energy Saving (%). Chronis et al. [10] provide no formal DP and report no quantitative evaluation metrics. Liu et al. [11] use a far weaker privacy budget ($\epsilon = 8$) and evaluate on non-smart-home tasks, making direct metric comparison inappropriate.

Two observations from Table IV are noteworthy. First, the proposed framework operates at a substantially stronger privacy guarantee ($\epsilon = 1.0$–2.0) than Liu et al. [11] ($\epsilon = 8$), yet still achieves competitive recommendation quality on a domain-specific task, demonstrating that the QLoRA-based approach is more privacy-efficient than the Gaussian DP-LoRA approach for this class of applications. Second, unlike Chronis et al. [10], who provide no quantitative evaluation, the proposed framework establishes a measurable and reproducible performance baseline for privacy-preserving LLM-based smart-home recommendations, filling a gap in the existing literature.

V. SYSTEM DEMONSTRATION AND OUTPUTS

In this study, the LLM-based recommendation system generates personalized energy-saving suggestions for smart home environments by considering multiple input parameters, including temperature, humidity, energy usage, and motion detection. Unlike centralized systems [16], which pose significant privacy risks due to centralized data storage, our approach enhances privacy by leveraging federated learning and differential privacy, ensuring sensitive data remains localized and reducing the risk of breaches. Compared to hybrid models [6], which offer improved personalization but suffer from moderate privacy and scalability limitations, our decentralized framework ensures both high privacy and scalability while maintaining efficient processing. Similarly, while IoT models [7] are highly scalable, they often compromise on privacy and personalization, challenges that our approach overcomes through LLM-driven recommendations integrated with secure decentralized learning techniques. By balancing privacy, scalability, personalization, and accuracy, our proposed system outperforms traditional models, providing a robust and secure solution for smart home automation.

Our smart home energy management system demonstrates real-time data visualization and provides AI-generated recommendations for energy optimization. The results are presented through a user-friendly dashboard and actionable suggestions.

A. Real-time Data Visualization

The system successfully captures and displays real-time data from various sensors, as shown in Figure 5:

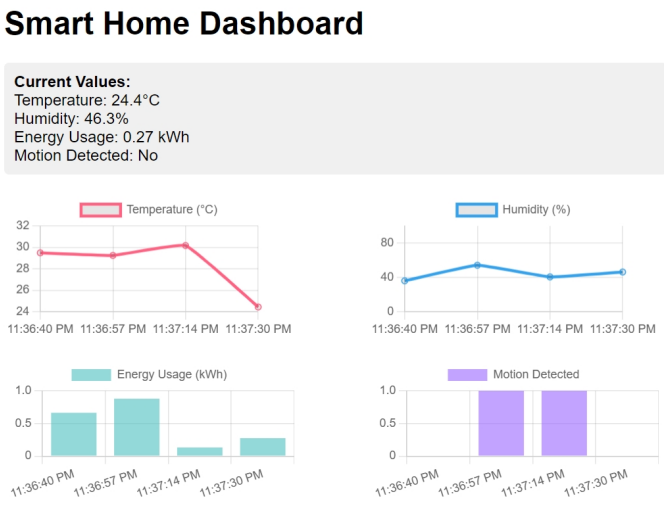


Fig. 5. Real-time data visualization

- *Temperature:* The graph shows a temperature range between 24°C and 30°C over the observed period, with a current temperature of 24.4°C.

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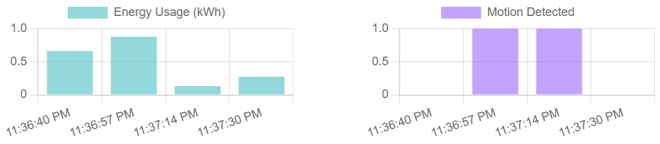
- *Humidity:* Humidity levels fluctuated between approximately 40% and 60%, with a current reading of 46.3%.
- *Energy Usage:* Energy consumption varied significantly, with peaks reaching nearly 1 kWh and lows near 0 kWh. The current energy usage is 0.27 kWh.
- *Motion Detection:* The system effectively tracked occupancy, showing periods of detected motion and no motion.

The dashboard's ability to display these metrics simultaneously allows users to correlate different factors affecting energy consumption.

B. AI-Generated Recommendations

In Figure 6, the system provides actionable, context-sensitive recommendations based on current sensor readings:

1) *Temperature Optimization:* The AI suggested reducing the thermostat temperature to 22°C from the current 24.4°C, estimating this would save energy without compromising comfort.
2) *Lighting Efficiency:* Noticing the light usage of 0.25 (on a scale of 0-1) despite no motion detection, the system recommended turning off lights in unoccupied rooms.



Recommendations

- **1. Reduce the thermostat temperature to 22°C to save energy.** The current temperature is 24.4°C, which is higher than the recommended thermostat setting of 22°C. Reducing the thermostat temperature will save energy without compromising comfort. **2. Turn off the lights if no one is in the room.** The light usage is currently 0.25 (scale 0-1), which indicates that the lights are on even though no one is in the room. Turning off the lights when no one is in the room will save energy.

Fig. 6. LLM Recommendation for efficient energy consumption

These results demonstrate the system's capability to monitor, analyze, and provide actionable insights for smart home energy management in real time. The AI-driven recommendations show potential for significant energy savings by addressing specific situations detected by the system.

VI. Conclusion and Future Work

This paper presented a privacy-preserving smart home recommendation framework that integrates LLaMA 3.2 3B, federated learning, and a dual-layer Laplacian differential privacy mechanism. By combining QLoRA-based parameter-efficient fine-tuning with a two-tier Home Knowledge server architecture, the framework delivers context-aware, personalized energy recommendations while ensuring that sensitive user data never leaves the local network. Quantitative evaluation demonstrates that meaningful recommendation quality is preserved across a range of privacy budgets ($\epsilon = 0.5$–5.0), and that the LLM-based approach substantially outperforms threshold-based rule systems on the same task.

Future work will focus on two directions. First, validation on real-world smart home sensor deployments will move the framework beyond simulated data and confirm its practical applicability. Second, the privacy-utility trade-off will be further optimized by exploring adaptive noise mechanisms that dynamically adjust ϵ based on data sensitivity, rather than applying a fixed budget across all devices. Together, these steps will strengthen both the empirical foundations and the practical deployability of privacy-preserving recommendation systems for smart home environments.

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